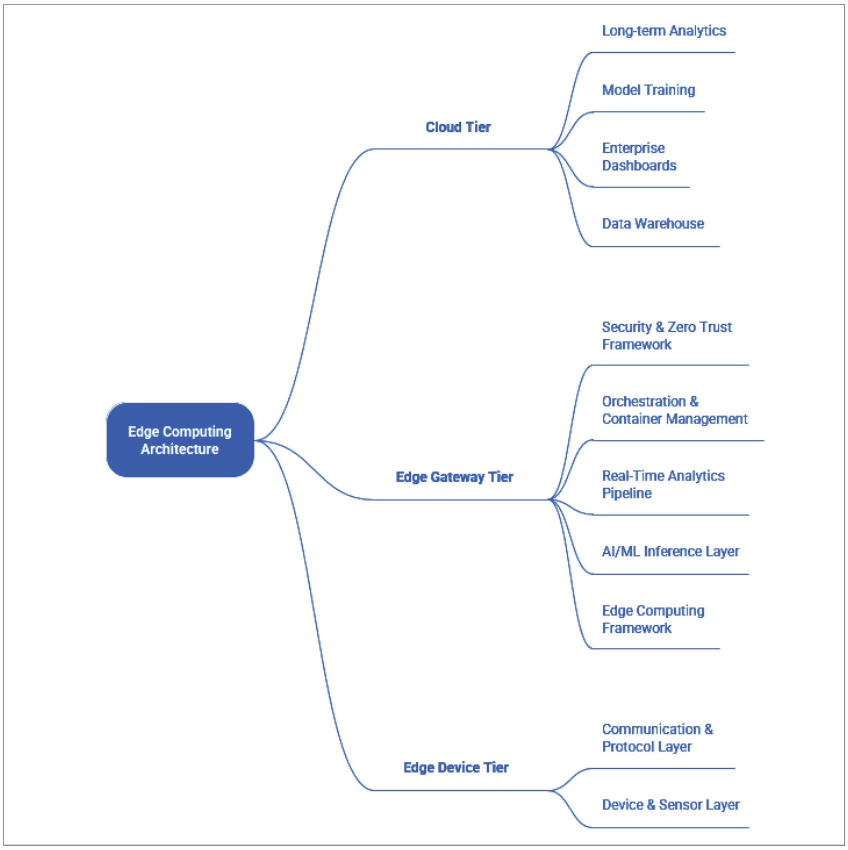

Figure 1: Edge computing architecture overview

Layer 4: AI/ML inference layer - the intelligence core: The AI inference layer evolves edge devices from simple data collectors into intelligent decision-makers. TensorFlow Lite is the leading framework for edge AI, optimising model sizes and accelerating inference through techniques like quantization and pruning. TensorFlow Lite for Microcontrollers enables complex neural networks on devices with under 50 kilobytes of memory, making it ideal for battery-powered sensors and wearables.

Layer 5: Real-time analytics pipeline - the intelligence amplifier: The analytics pipeline aggregates insights across edge devices, revealing patterns not visible in isolated nodes. Apache Kafka acts as a distributed commit log, while Spark Streaming processes events in micro-batches, enabling fleet-wide anomaly detection and adaptive control strategies. Alternative architectures, such as InfluxDB and Grafana, focus on time-series data, optimising storage and queries for metrics-driven applications.

Layer 6: Orchestration and container management - the operational foundation: K3s simplifies Kubernetes for edge environments, offering a lightweight footprint that enables the management of containerised workloads on resource-constrained devices. It ensures consistent deployments across diverse infrastructures, supports reliable rollouts and rollbacks, and manages the complexities of distributed systems. Service mesh technologies, such as Linkerd, enhance security and observability. Container orchestration allows sophisticated deployment strategies, ensuring systems run reliably over time with minimal intervention.

Layer 7: Security framework - the trust foundation: Edge deployments face unique security challenges due to physical accessibility and hostile network environments. Zero trust architecture and robust security measures are essential to protect these systems effectively. Network segmentation isolates device networks from enterprise networks and the public internet, utilising secure tunnelling protocols like WireGuard to provide encrypted communication channels with minimal overhead.

This layered architecture enables teams to work on different layers independently while maintaining well-defined interfaces. Open source components at every layer eliminate vendor lock-in while providing production-grade capabilities developed and validated by global communities.

However, this sophistication comes with complexity. The learning curve spans multiple domains, including embedded systems and networking, as well as distributed systems and machine learning. Integration challenges arise at layer boundaries, requiring careful attention to data formats and error handling.

Real-world application: Intelligent retail supply chain

The retail industry faces relentless pressure to optimise inventory, reduce waste, and deliver seamless customer experiences while operating on razor-thin margins. Traditional supply chain management relies on batch processing, periodic inventory counts, and reactive responses to stockouts or overstocking situations. Edge AI transforms this reactive model into a proactive, continuously optimising system that anticipates problems before they impact operations and customers.

The challenge: Complexity at scale: Modern retail supply chains operate with extraordinary complexity. Distribution centres process thousands of SKUs daily, each with unique handling requirements, shelf-life constraints, and demand patterns.